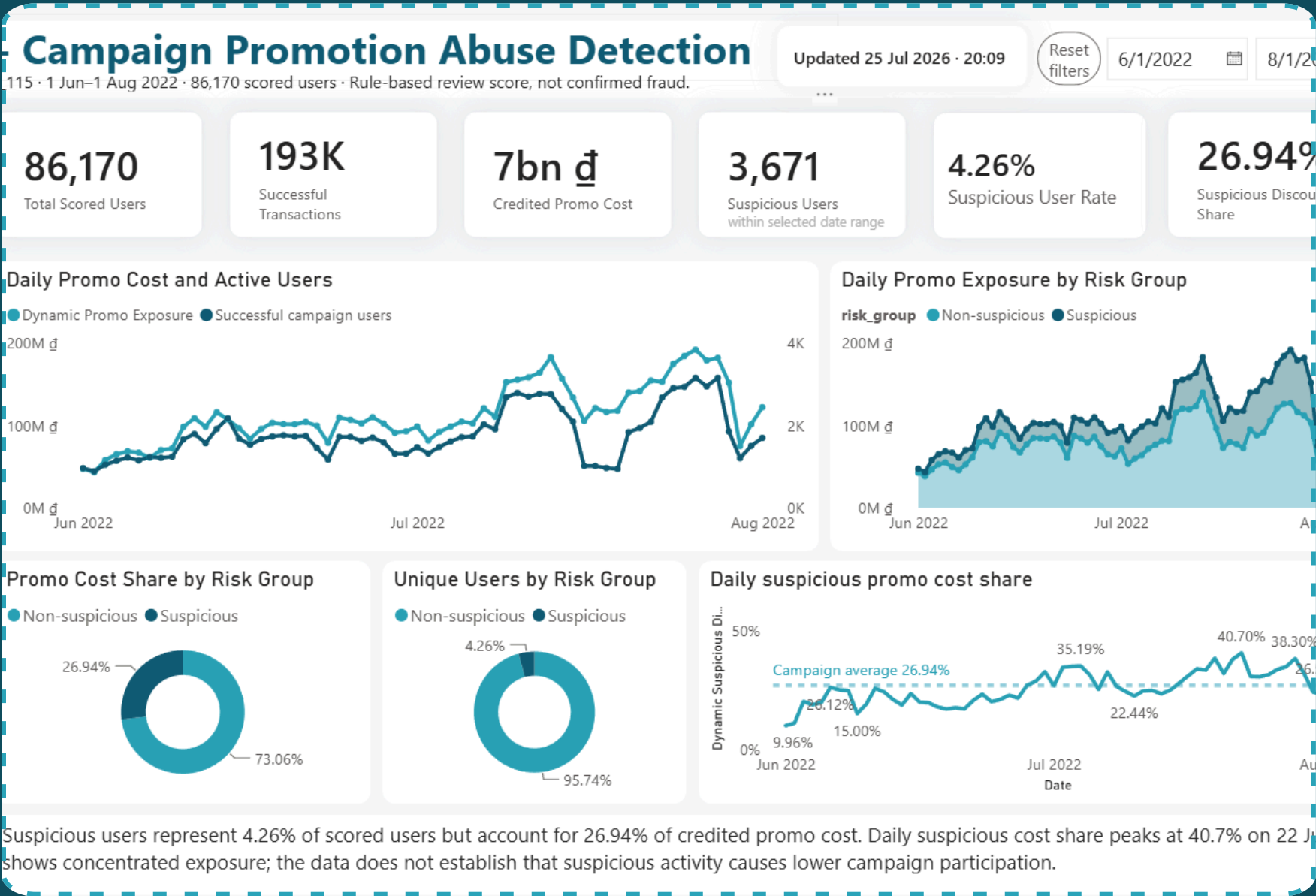


Promotion-abuse analytics

Turning 7.0bn đ of campaign activity into an explainable review strategy

SQL → Python → Power BI



Rule-based review score — not confirmed fraud

The decision: where review effort creates the most value



2.31%
review workload under the
recommended rule

23.89%
promo cost captured by that rule

Review first
validate in shadow mode before any
blocking

A small user segment carries disproportionate cost, but the correct action is prioritized review—not automated enforcement.

A three-phase path from campaign data to a review decision

3 phases

context → evidence → decision

A 6–8 minute stakeholder story focused on business judgment.

- 01

Context and scope

Dataset origin, campaign scale and the decision problem.
- 02

Evidence and benchmarks

Concentration, user signals and seven rule scenarios.
- 03

Decision and delivery

Recommended action, dashboard product and evidence limits.

Three questions turn campaign leakage into a decision

7.0bn ₪

credited promo cost inside the selected campaign

73 promotions · 254,309 transaction rows ·
90,555 campaign users

01

Where is promo cost concentrated?

Find the promotions and merchant categories that dominate exposure.

02

Which users warrant review—and why?

Use explainable signals instead of a black-box label.

03

Which rule balances coverage and workload?

Choose a practical review policy, not the broadest possible rule.

The campaign was chosen by a priority scan, not assumed

	campaignCode	campaign_id_count	transaction_rows	unique_users
1	ZPI_220801_115	73	254309	90555
2	DGS_230802_738	4	19474	18115
3	ZPI_230808_525	56	232246	21676
4	ZPI_230601_511	2	22084	22084
5	ZPI_230518_498	1	11455	1224
6	ZPI_230725_599	7	5462	3147
7	MKT_230607_538	1	6967	4704
8	DGS_230701_575	1	1171	1033
9	DGS_230707_603	8	4235	3298
10	ZPI_230712_644	1	1641	1409

Top 1 of many
campaign ranked by combined scale and credited promo cost

73
sub-campaigns inside the selected campaign

ZPI_220801_115
selected for deep-dive scoring

Campaign priority scan — all campaigns ranked

Rule-based review score — not confirmed fraud

254K transactions narrow to an 86K-user review population

254,309

transaction rows in campaign
ZPI_220801_115

1 Jun–1 Aug 2022 · 90,555 users · 73
promotions

01

Campaign activity

193,295 successful rows · 7.004bn ₺ credited promo cost.

02

Analytical output

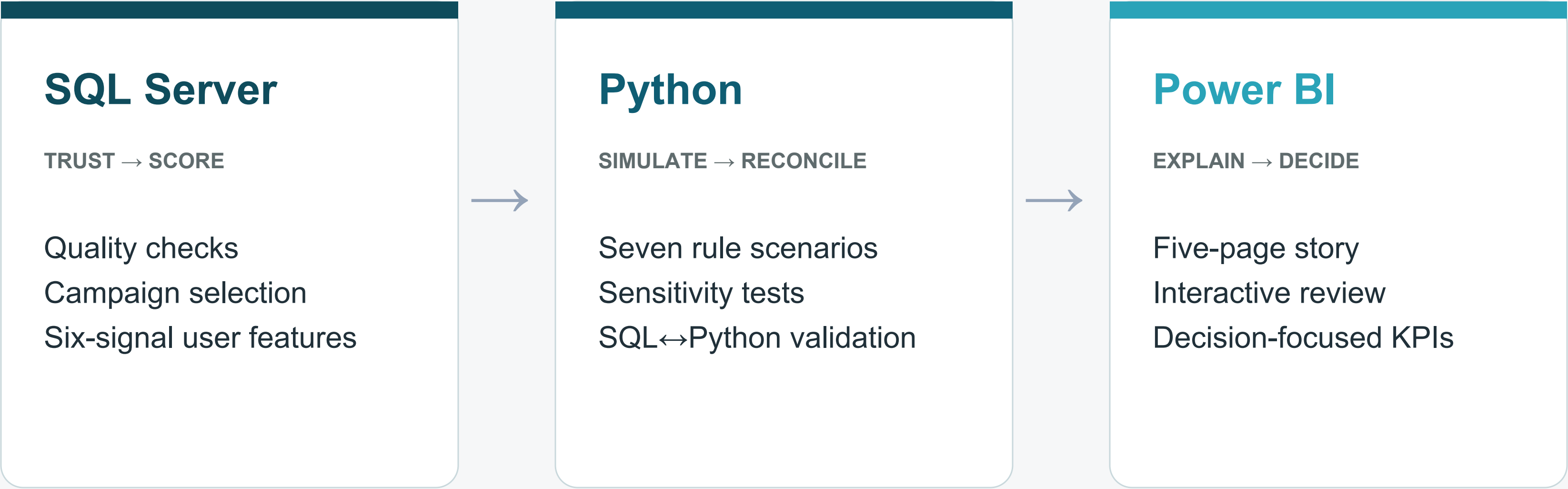
86,170 scored users · 3,671 prioritized for review.

03

Evidence boundary

No confirmed fraud labels; scores rank review priority.

SQL established trust; Python tested options; Power BI carried the decision



Trust gate 20 SQL↔Python outputs matched; slide 14 explains what this validates.

A six-signal score makes the review logic explainable

Behavioral evidence

- Immediate discount after joining
- High credited campaign discount
- Referral farming
- Shared-device usage
- Shared-IP usage
- Reciprocal transfer loops

0–2 points per signal

0–11

suspicion score

LOW 0–2

REVIEW 3–4

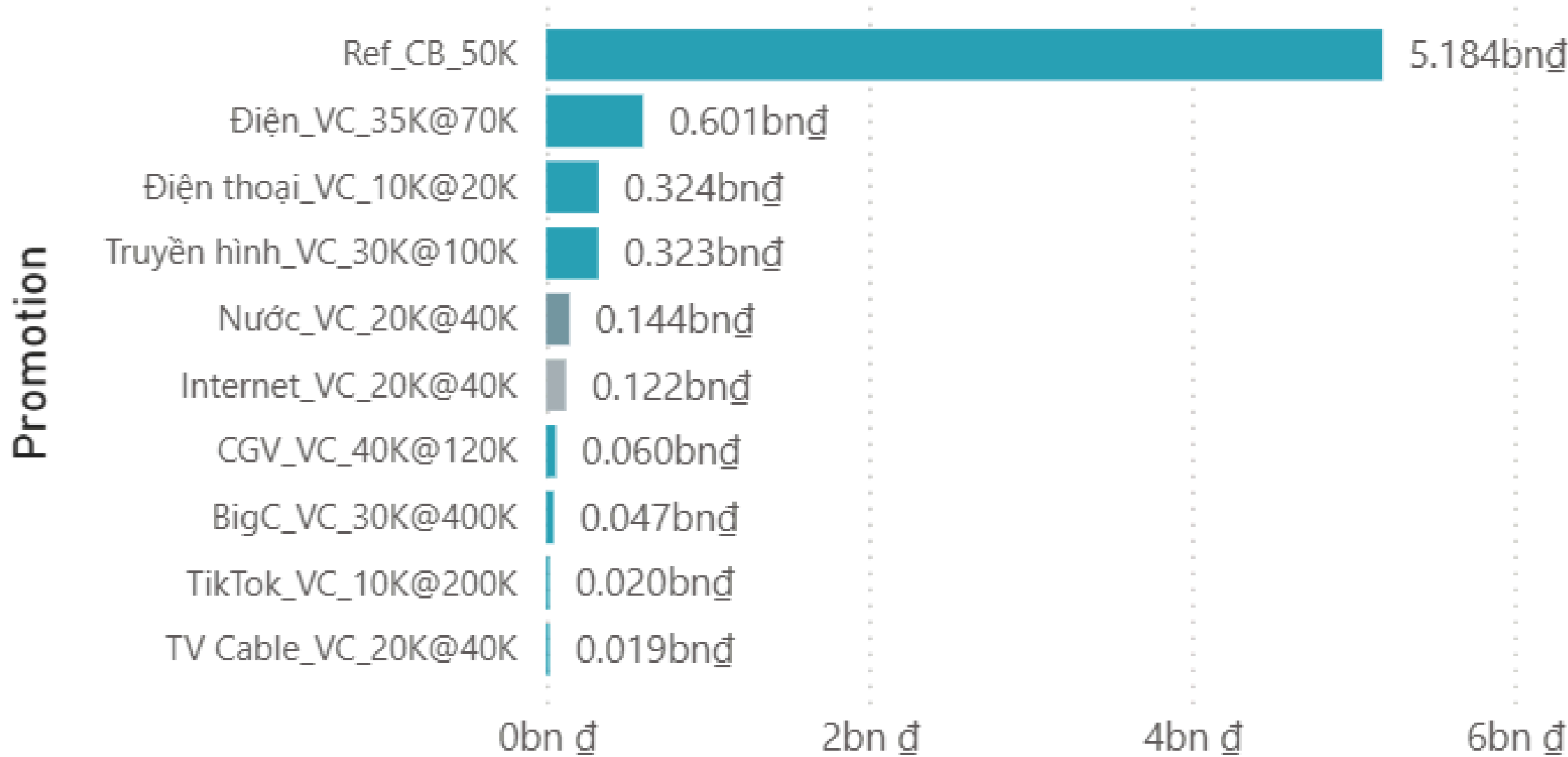
MEDIUM 5–6

HIGH 7+

The score prioritizes review; it does not prove abuse.

One promotion accounts for 74% of credited promo cost

Credited promo cost by promotion



5.18bn đ

credited promo cost in Ref_CB_50K

74.02%

of campaign credited promo cost

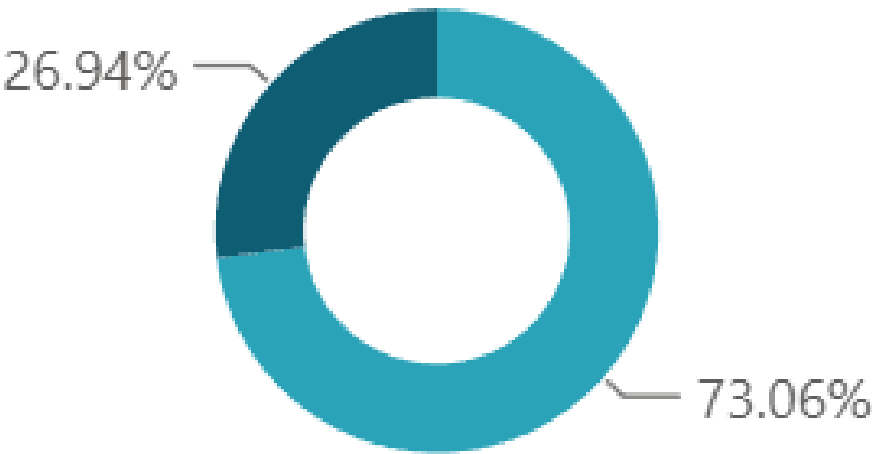
Implication

Monitoring can focus on the dominant promotion instead of treating all 73 promotions equally.

A small user segment carries a disproportionate share of cost

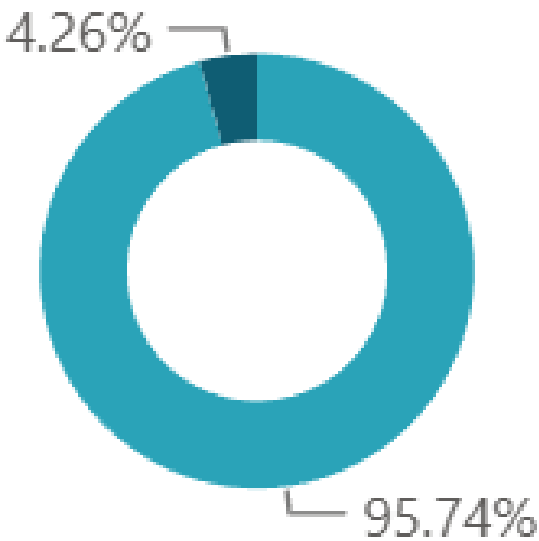
Promo Cost Share by Risk Group

● Non-suspicious ● Suspicious



Unique Users by Risk Group

● Non-suspicious ● Suspicious



6.3x

cost concentration versus population share

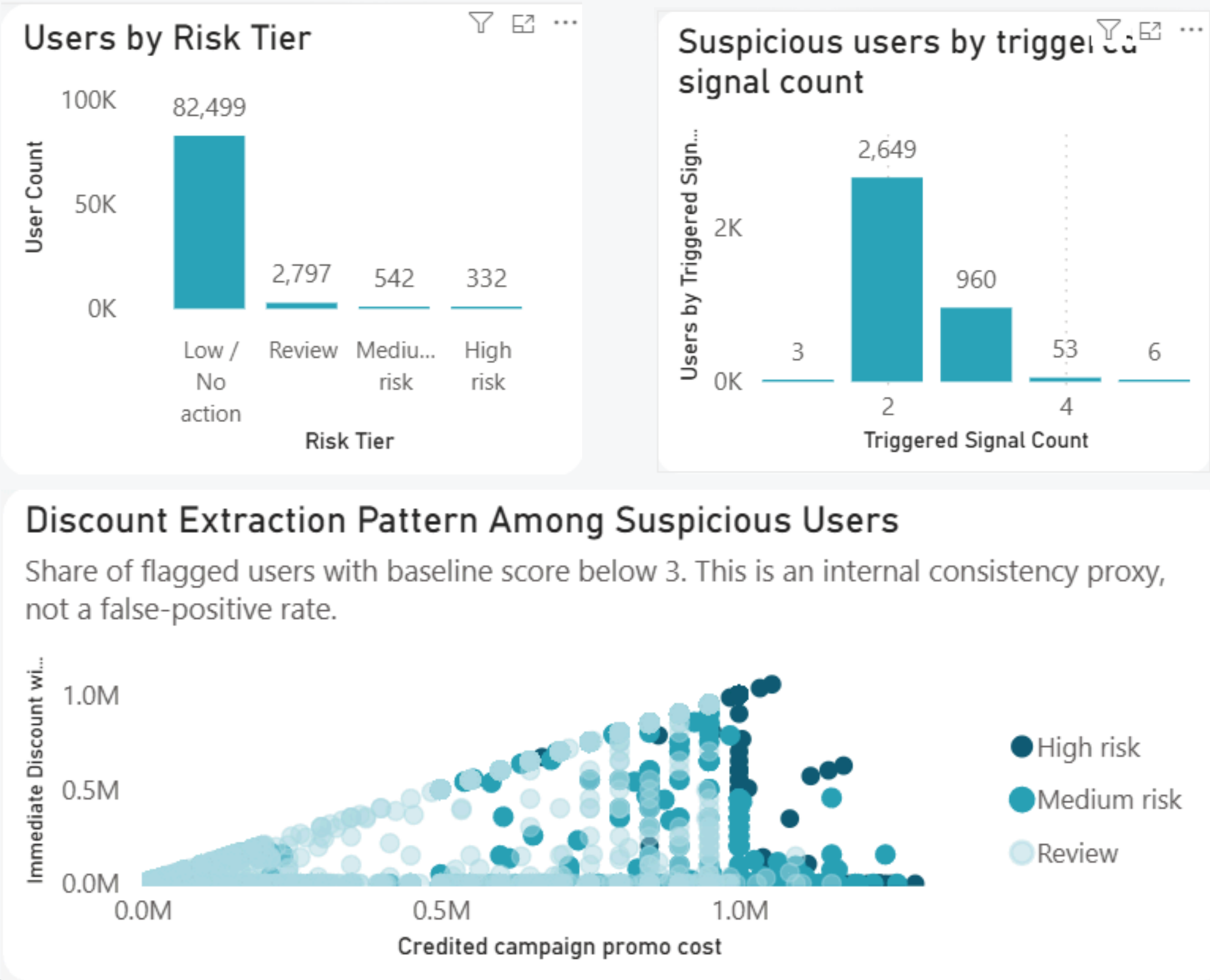
1.89bn đ

credited promo cost linked to suspicious users

Peak daily share: 40.7% on 22 Jul

Concentrated exposure—not evidence of causation.

Suspicious users are almost always multi-signal cases



99.9%

trigger at least two current scoring signals

27.8%

trigger three or more signals

Most prevalent signals

Credited discount · transfer loop · shared IP

Seven rules define the trade-off between coverage and workload

All seven rule scenarios — users flagged, cost captured, over-flagging

Share of flagged users with a baseline score below 3. This is an internal consistency proxy, not a confirmed false-positive rate.

Rule	#Users	Promo cost captured	Low-score proxy share	Recommended Use
Medium balanced rule	1994	23.89	0.00	Candidate for controlled blocking test or high-priority review
New-account extraction rule	1193	13.83	0.00	Candidate for controlled blocking test or high-priority review
Referral farming rule	1080	14.45	0.00	Candidate for controlled blocking test or high-priority review
Broad monitoring rule	19332	68.02	83.47	Monitoring only; too broad for direct blocking
Shared device rule	23	0.06	13.04	Supporting rule; combine with other signals
Strict multi-signal rule	4	0.05	0.00	Supporting rule; combine with other signals
Transfer-loop review rule	537	2.57	0.00	Supporting rule; combine with other signals

Broad — too disruptive

68.02% cost captured
but 83.47% of flagged users score below 3

Balanced — recommended

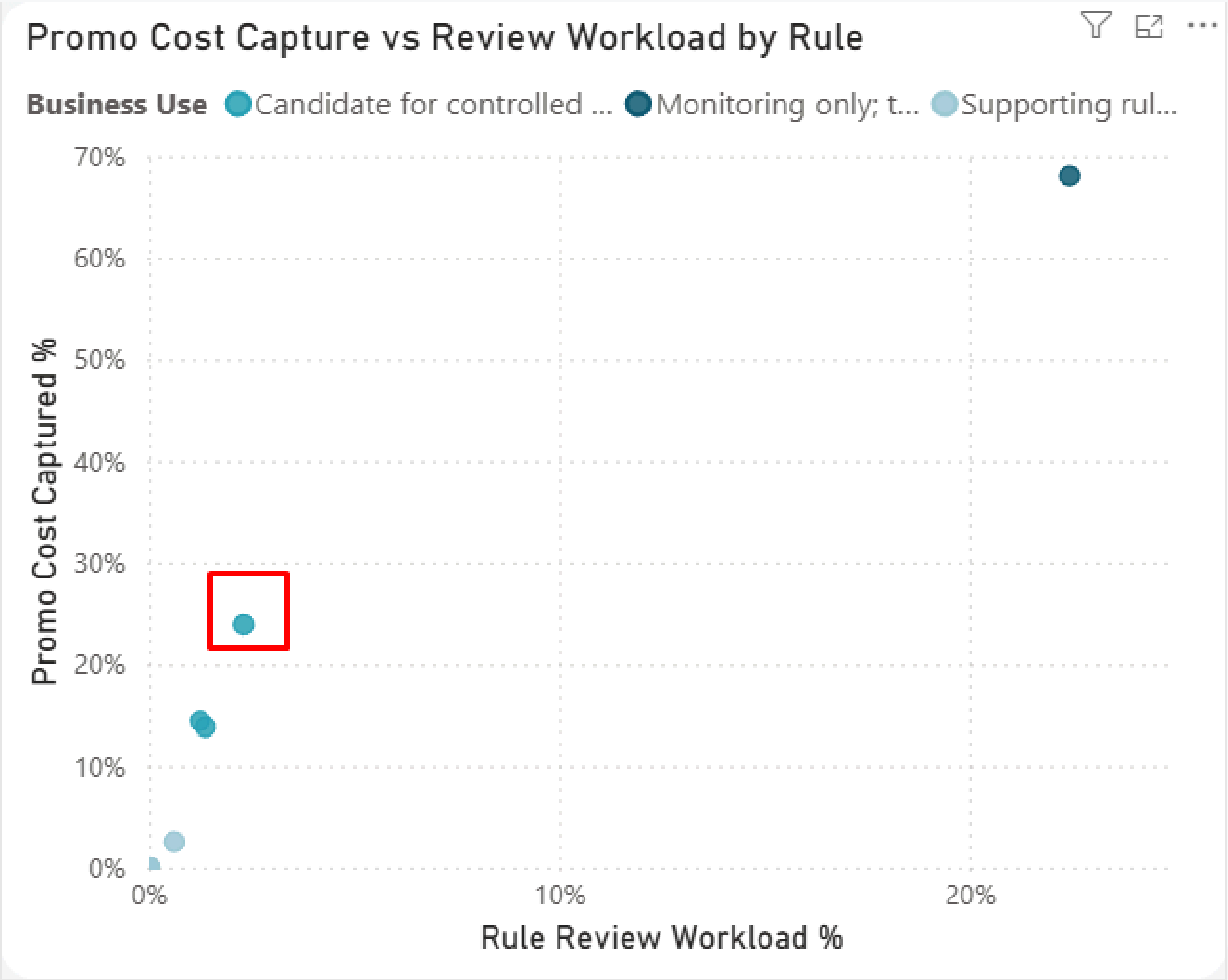
23.89% cost captured at 2.31% workload
0% low-score proxy impact

Narrow — too little reach

1.25% to 0.005% workload
only 14.45% down to 0.05% captured

Only the medium-balanced rule pairs meaningful coverage with near-zero over-flagging.

The medium-balanced rule delivers the clearest trade-off



The red marker shows the medium-balanced rule — high capture at low workload.

1,994

users flagged

2.31%

review workload

23.89%

promo cost captured

54.32% of current score-3+ users captured · 0% low-score proxy impact

Every flagged user comes with a readable reason

userID	suspicion_score	risk_tier	reason
U052491	11	High risk	high immediate campaign discount within 0-1 day: 1055000, high credited campaign discount: 1055000, high referral count: 23, sha
U000648	9	High risk	high credited campaign discount: 1295000, high referral count: 20, shared device used by 5 users, shared IP used by 36 users, recip
U078015	9	High risk	high credited campaign discount: 1265000, high referral count: 20, shared device used by 5 users, shared IP used by 36 users, recip
U000108	9	High risk	high credited campaign discount: 1155000, high referral count: 20, shared device used by 5 users, shared IP used by 36 users, recip
U028616	9	High risk	high immediate campaign discount within 0-1 day: 1000000, high credited campaign discount: 1000000, high referral count: 20, sha
U077555	9	High risk	high credited campaign discount: 1000000, high referral count: 20, shared device used by 5 users, shared IP used by 36 users, recip
U037271	8	High risk	high credited campaign discount: 1255000, high referral count: 20, shared IP used by 36 users, reciprocal transfer loop with 27 part
U083314	8	High risk	high credited campaign discount: 1245000, high referral count: 20, shared IP used by 36 users, reciprocal transfer loop with 14 part
U000534	8	High risk	high credited campaign discount: 1235000, high referral count: 20, shared IP used by 36 users, reciprocal transfer loop with 22 part
U035519	8	High risk	high credited campaign discount: 1235000, high referral count: 20, shared IP used by 36 users, reciprocal transfer loop with 9 partn
U064212	8	High risk	high credited campaign discount: 1225000, high referral count: 20, shared IP used by 36 users, reciprocal transfer loop with 19 part
U045411	8	High risk	high credited campaign discount: 1215000, high referral count: 20, shared IP used by 36 users, reciprocal transfer loop with 22 part
U041660	8	High risk	high credited campaign discount: 1185000, high referral count: 22, shared IP used by 36 users, reciprocal transfer loop with 38 part
U046497	8	High risk	high immediate campaign discount within 0-1 day: 625000, high credited campaign discount: 1175000, high referral count: 20, recip
U053864	8	High risk	high credited campaign discount: 1175000, high referral count: 20, shared IP used by 24 users, reciprocal transfer loop with 7 partn
U085705	8	High risk	high immediate campaign discount within 0-1 day: 570000, high credited campaign discount: 1120000, high referral count: 22, recip
U027357	8	High risk	high credited campaign discount: 1070000, high referral count: 21, shared device used by 5 users, reciprocal transfer loop with 20 p
U033592	8	High risk	high credited campaign discount: 1065000, high referral count: 22, shared device used by 11 users, reciprocal transfer loop with 1 p
U008716	8	High risk	high immediate campaign discount within 0-1 day: 1035000, high credited campaign discount: 1035000, high referral count: 22,
U088992	8	High risk	high immediate campaign discount within 0-1 day: 765000, high credited campaign discount: 1005000, high referral count: 28, recip
U000650	8	High risk	high immediate campaign discount within 0-1 day: 1000000, high credited campaign discount: 1000000, high referral count: 20,
U000711	8	High risk	high immediate campaign discount within 0-1 day: 1000000, high credited campaign discount: 1000000, high referral count: 20,
U000745	8	High risk	high immediate campaign discount within 0-1 day: 1000000, high credited campaign discount: 1000000, high referral count: 20,
U001472	8	High risk	high immediate campaign discount within 0-1 day: 1000000, high credited campaign discount: 1000000, high referral count: 21,
U001490	8	High risk	high immediate campaign discount within 0-1 day: 1000000, high credited campaign discount: 1000000, high referral count: 20,
U001801	8	High risk	high immediate campaign discount within 0-1 day: 1000000, high credited campaign discount: 1000000, high referral count: 21,
U002046	8	High risk	high immediate campaign discount within 0-1 day: 1000000, high credited campaign discount: 1000000, high referral count: 21,
U002539	8	High risk	high immediate campaign discount within 0-1 day: 1000000, high credited campaign discount: 1000000, high referral count: 20,

Review queue — score, signals and reason

What a reviewer receives

- A suspicion score from 0 to 11
- The specific signals that fired
- A plain-language reason string
- A risk tier that sets review priority

The output is auditable at the individual level, which is what makes review defensible.

Use the rule to learn before using it to block



Run in shadow mode

Score users without changing the customer experience.



Review and label

Sample flagged and unflagged users; record investigation outcomes.



Decide the threshold

Measure precision, recall, workload and customer harm before automation.

RECOMMENDED RULE AT A GLANCE

≈1.7bn ₪
promo cost placed under review

2.31%
of scored users reviewed

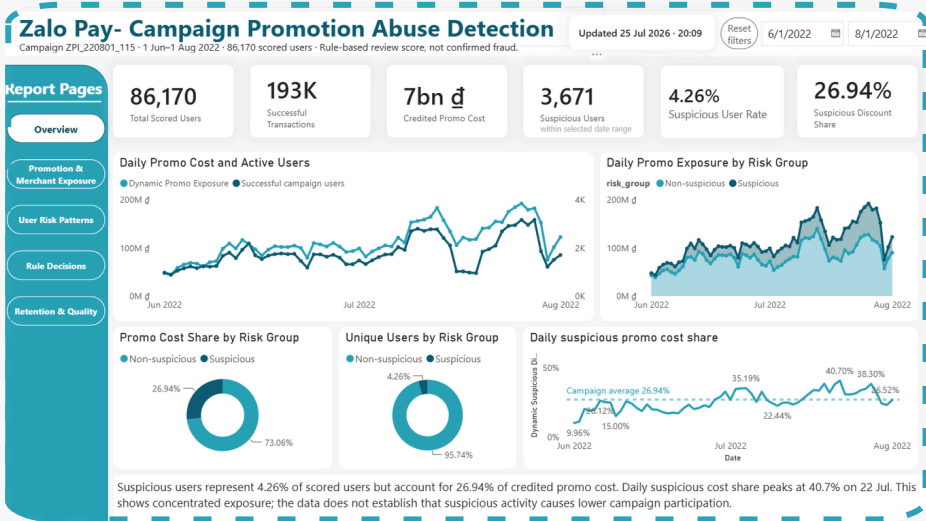
0%
low-score proxy impact

Exposure surfaced for review — not recovered value.

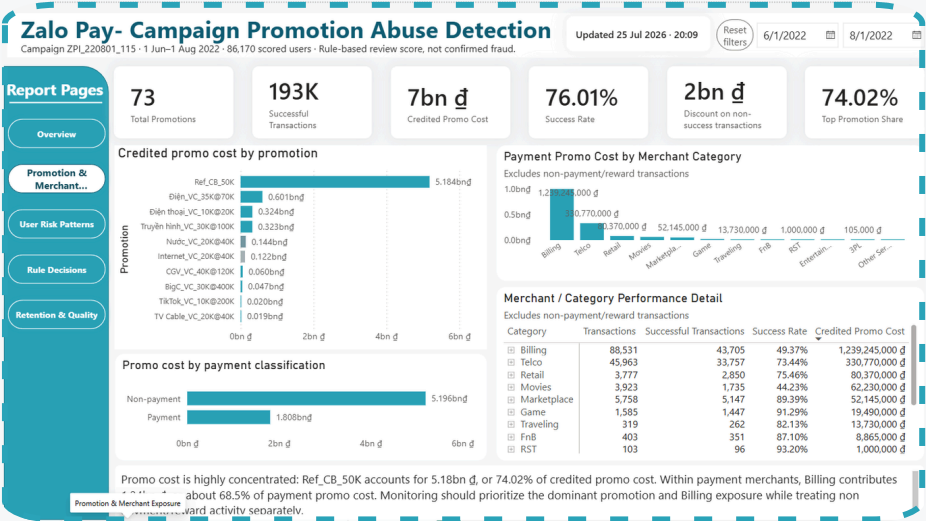
Immediate recommendation

Prioritize the dominant promotion and route medium-balanced-rule cases to high-priority manual review.

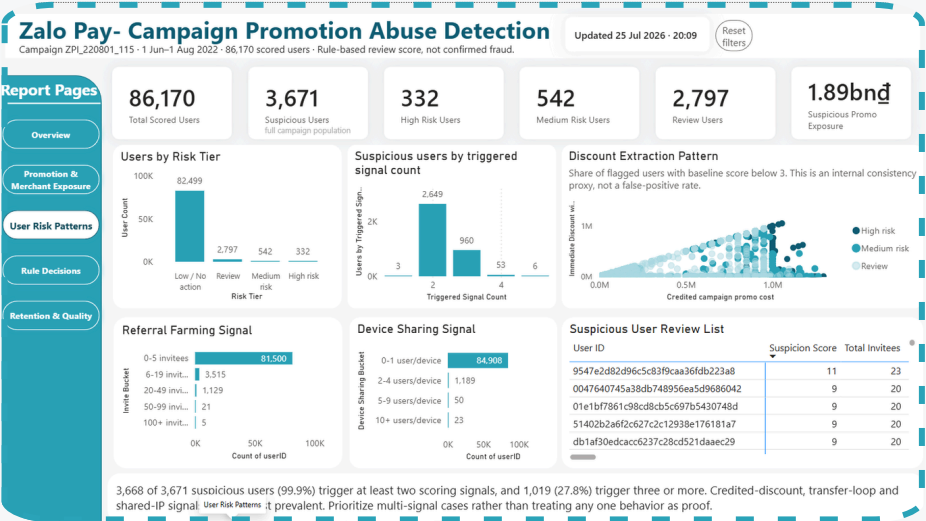
The dashboard turns analysis into a five-page decision path



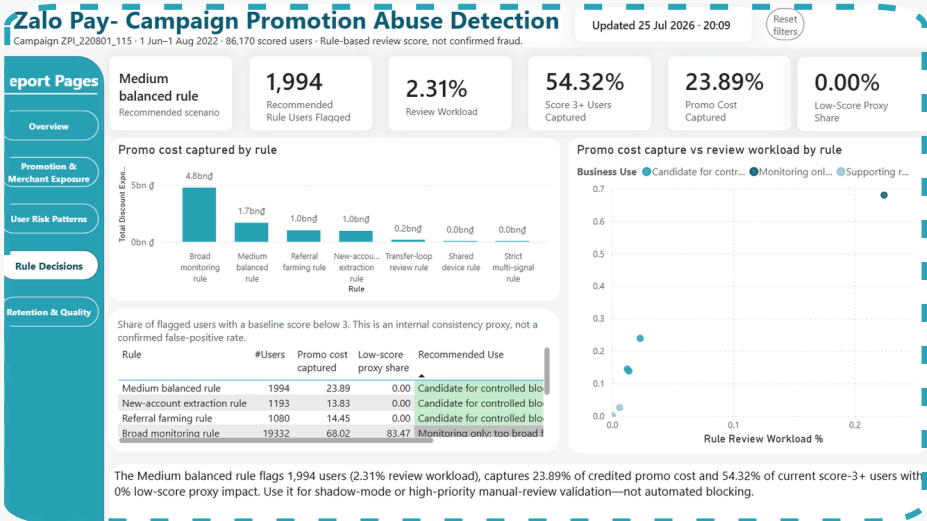
Overview
Scale and imbalance



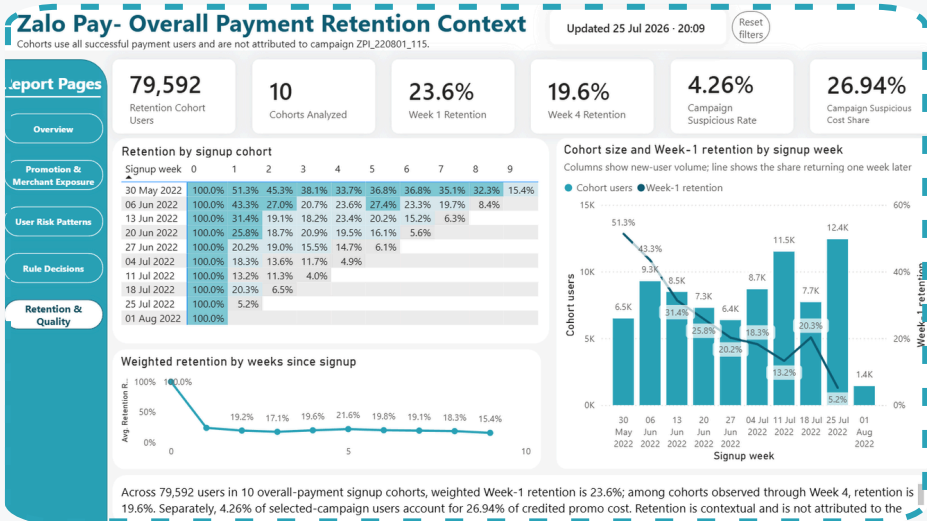
Promotion & Merchant
Concentration



User Risk Patterns
Behavioral evidence



Rule Decisions
Coverage vs workload



Retention & Quality
Context and caveats

Problem → exposure → behavior → decision → broader retention context

The evidence is reconciled—but the outcome is not labeled

20 / 20

SQL↔Python reconciliation checks
matched

A transfer-loop self-join bug was
corrected from an inflated 500,057
count to a distinct-partner maximum of
158.

What the project supports

- ✓ Explainable review prioritization
- ✓ Cost-exposure concentration
- ✓ Rule workload trade-offs
- ✓ Reproducible SQL↔Python checks

What it does not prove

- ✗ Confirmed fraud
- ✗ A true false-positive rate
- ✗ Causal campaign retention
- ✗ Readiness for automated blocking

Next evidence to collect: manual-review outcomes,
network/ring labels and post-action customer impact.

Focus review now; automate only with evidence

Concentrated exposure and explainable signals make a targeted review strategy possible.

74.02%
cost focus

1,994
users flagged

Shadow-mode
test recommended

SQL → Power BI
pipeline delivery

LIVE DASHBOARD

youtu.be/_UaicJaSGOs

PROJECT REPOSITORY

<https://github.com/kath-pahotu/zalopay-promo-abuse-detection>

Decision: prioritize Ref_CB_50K and medium-balanced cases; collect investigation labels before any blocking.

Phan Hoai Thu · Data analyst portfolio